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Business Method Innovation in U.S. Manufacturing and Trade

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Abstract. *Problem definition:* In what kinds of business method innovation do firms in the manufacturing and trade sectors engage? Does engaging in business method innovation create value for these firms? The present paper answers these questions using empirical evidence. *Methodology/results:* Using text analysis of business method patents, we show that business method innovation in the U.S. manufacturing and trade sectors is aimed primarily at improving the business operations that support the sales of tangible products—that is, how the firm targets customers, manages product delivery, or enhances the product through service offerings. We then evaluate the effect of having business method innovation, as evidenced by patents, on a firm's value. Leveraging the exogenous shock of the *State Street* ruling, which first recognized business methods as a patentable category of innovation, we identify a set of firms that possess business method patents and a matched set of comparable firms without such patents. Then, using a difference-in-differences with firm fixed effects model on the matched sample, we show that the valuation of the former set of firms increased by 9% after *State Street*, as measured by Tobin's q . We further show that (1) business method innovators in the manufacturing sector gained a 7% increase, whereas business method innovators in the trade sectors gained a 25% increase; and (2) only firms with broader innovation scope—that is, business method innovations covering the range of customer targeting, product delivery, and service support of products—experienced a significant (18%) value bump. *Managerial implications:* This research provides evidence that business method innovation in the manufacturing and trade sectors primarily involves innovating in business operations that support product sales. Our work also provides empirical support for the proposition that engaging in business method innovation drives manufacturing and trade firms' market performance.

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Keywords: business methods • Tobin's q • patents • manufacturing • distribution • retail

Introduction

Business models, or blueprints of how a firm creates and delivers value to customers, have received much attention in the digital age (Girotra and Netessine 2014, Plambeck and Ramdas 2020). Although the concept of a business model remains elusive (Massa and Tucci 2013) with various conceptualizations (Zott et al. 2011), scholars tend to agree that a business model is a “system level concept, centered on activities and focusing on value” (Zott et al. 2011, p. 1037). Correspondingly, a business model draws attention to how an organization arranges its system of activities for value creation.

For firms in manufacturing and trade—that is, firms involved in manufacturing and distributing tangible

goods—the term “business model” attains a more specific meaning.¹ For these firms, a business model refers to the business operations supporting and enhancing the core product that can happen “both before and after sale” and can be segmented into how a “tangible product is marketed, delivered to customers, or enhanced with services” (Netessine and Girotra 2014, p. 12). Furthermore, just as a firm can engage in product and process technology innovation to improve elements of its product or manufacturing process, it can also engage in *business method innovation* to improve elements of its business model. Because business method innovation for firms in manufacturing and trade involves reconfiguring business activities to support product sales, it is distinct from product and process technology innovation.

Consider the example of Dell Inc., which has a make-to-order business model alongside its product-market strategy of selling an array of highly customizable computer products (Pisano and Teece 2007). Not only does Dell have a large portfolio of computer product and manufacturing technology patents, but it also has several patented business methods to improve the functioning of its make-to-order business model (Rivette and Kline 2000). These patented business methods include innovations to aid marketing and consumer choice (e.g., US 6578013 to recommend products that are configurationally consistent with a client's existing products); innovations to smooth product delivery (e.g., US 7756760 to determine inventory shortage and delay costs); and innovations that enhance support services (e.g., US 7373307 to determine an optimal warranty configuration for a computer product). Notably, all these business method innovations concern different aspects of Dell's business model.

Assessing the scope and impact of business method innovation has been difficult, however, because it is challenging to empirically observe such activities within the firm. Significantly, it was not until the late 1990s that firms could patent their business method innovations. In fact, innovations on “ways of doing business” (including, but not limited to, innovations supporting the sales of tangible products with which we are concerned in the present paper) were not recognized as patentable until the 1998 decision, *State Street Bank and Trust Co., Inc. v. Signature Financial Group, Inc.*, 149 F.3d 1368 (Fed. Cir. 1998). This decision led to a sharp jump in the number of business method patent applications that the U.S. Patent and Trademark Office (USPTO) received (Meurer 2002, USPTO 2017).

Since *State Street*, the few empirical studies to examine business method innovations have tended to focus on the financial and information sectors (Lerner 2002, Zott and Amit 2008, Hall and MacGarvie 2010, Desyllas and Sako 2013). Because these sectors primarily deal with intangible goods, their business method innovations can often be characterized as financial or software product innovations (Hunt 2010). Thus, for example, Lerner (2002) equates business method patents in the finance sector as indicative of “new product development” (p. 908) of financial products. Curiously, in sectors involving the manufacture, distribution, and retail of physical goods, where we expect business method innovations supporting the sales, delivery, and servicing of physical products to dominate, the economic significance of business method innovations has been overlooked. This oversight is particularly notable given that firms in these sectors account for nearly a third of all business method patents issued.

We fill this gap in the literature on business methods by asking two related research questions. First, in

what kinds of business method innovation do firms in manufacturing and trade (i.e., wholesale, retail, and transportation/warehousing) engage? In answering this question, we provide a detailed descriptive analysis of firms' business method innovation, as well its alignment with a conceptual typology of the different components of the business model (Girotra and Netessine 2014). Second, does engaging in business method innovation create value for firms in manufacturing and trade? Conceptual arguments offer conflicting guidance about the potential value of business method innovations to firms in these sectors. On the one hand, by engaging in business method innovation, firms can help identify novel ways to deliver value to customers. On the other, such engagement requires costly experimentation and reconfiguration of the firm's processes (Sosna et al. 2010) and can draw scarce resources and management attention away from product and manufacturing process innovations (Visnjic and Van Looy 2013). Business method patents represent discrete and documented efforts by firms to improve their business model. They allow us to answer both questions by exposing the content of a firm's business method innovation and by providing a way to link such innovation to financial performance.

As to the first research question, we analyzed and classified the abstracts of the business method patents granted to firms in the U.S. manufacturing and trade sectors. We show that most business method patents in these sectors fall into one of three groups: (1) those finding new ways to target, reach, and transact with customers (which we label “customer targeting”); (2) those enhancing operations to facilitate the delivery of products (“product delivery”); and (3) those determining how support services linked to a product should be configured, tracked, and charged (“support services”). We validated our classification in two ways: (1) using *Doc2Vec* (a text analysis approach that maps texts to a vector space whereby texts with similar meanings are put closer together; Le and Mikolov 2014), we show that our classification puts innovations that are closer in meaning in the same group; and (2) relying on online observers gathered from Amazon's Mechanical Turk (MTurk), we show that our classification is consistent with the groupings that MTurk observers generate.

Second, we examine the economic value of engaging in business method innovation. Using comprehensive patent data from 1986 to 2008, we leverage the exogenous shock of the 1998 *State Street* ruling, which unexpectedly opened the door for patenting business methods, to present new empirical evidence that establishes the economic significance of business method innovation to publicly traded U.S. firms in the manufacturing and trade sectors. Using a stringent matching approach followed by a difference-in-differences (DID) with panel

fixed effects model to exploit the *State Street* shock, we show that firms engaged in business method innovation (evidenced by obtaining at least one business method patent) experienced about a 9% bump in market value (measured by Tobin's q) over comparable firms without any patent grants in the post-*State Street* period. Our analysis is robust to various specifications and presents evidence that *State Street* resulted in business method innovators in the manufacturing and trade sectors receiving an economically and statistically significant value bump relative to nonbusiness method innovators. Because we compare business method innovators with a matched set of nonbusiness method innovators that have similar levels of research expenditure and patenting output, our results imply that firms in these sectors should explore adding business methods to their portfolio of product or process innovations. Thus, in addition to equipping the company with a great product lineup, a business model undergirded by innovative business methods provides a significant source of competitive advantage (Osterwalder and Pigneur 2010, Netessine and Girotra 2014).

Two additional analyses provide more detailed insights into how business method innovations create value. First, in a split sample analysis, we document that business method innovators in the manufacturing sector gained a 7% bump in market value after *State Street*, as opposed to a 25% bump for business method innovators in the trade sectors (i.e., the wholesale, retail, and transportation/warehousing industries). The 25% bump for firms in the trade sectors provides strong evidence that business method innovations are crucial for market value growth in these sectors. Stated differently, a large penalty is imposed for ignoring business method innovation if a firm's core activity is product sales and delivery. By contrast, the much smaller 7% bump in value for manufacturing firms may point to difficulties they face when balancing resources across different innovation activities (e.g., product, manufacturing process, and business method innovation). Yet even this more modest gain for manufacturers suggests that business method innovations provide important opportunities that should not be ignored.

Finally, linking to the classification developed for our first research question, we show that only firms with business method innovations across all application domains (customer targeting, product delivery, and support services) see a significant value bump after *State Street*. By contrast, firms with business method innovations that do not cover all three domains fail to experience such a value bump. In short, we observe significant economies of scope if firms can improve multiple components of their business model. We therefore provide the first empirical evidence to support existing theories about business models (Amit and Zott 2001, Girotra and

Netessine 2013) that suggest firms' ability to deliver value depends not just on innovating specific and isolated business methods but also on creating alignment and cohesion among various and improved business methods.

Business Method Innovations and Firm Value

Defining Business Method Innovation and Its Relation to the Business Model

Business method innovations or innovations on "ways of doing business" can be difficult to define when studied in the abstract and without a specific set of industries in mind. By focusing on industries involved in manufacturing and delivering physical products, one can more clearly distinguish between innovation activities focused on improving the product itself versus improving the business methods that support the product's sales (Netessine and Girotra 2014). On the one hand, product innovations and/or manufacturing technology innovations directly improve product attributes, such as performance, cost, or quality, and thus directly support a firm's product-market strategy (Zott and Amit 2008). By contrast, business method innovations create novel improvements in how firms market, deliver, and provide services related to such products. In this way, business method innovation directly supports the firm's "business model"—as Netessine and Girotra (2014) have used the term.

The Business Model Canvas (Osterwalder and Pigneur 2010), which has emerged as a popular practitioner tool for business model design, helps illustrate the different components of a business model. Osterwalder and Pigneur (2010) decompose a business model into nine elements, which can be grouped into three components describing (1) a firm's customer identification and targeting strategy (segments, channels, relationships, and value proposition), (2) its organization of operations (activities, resources, and partners), and (3) its finances (cost and revenue). The Business Model Canvas, however, is a generic tool and is not designed with a specific set of industries in mind.

Netessine and Girotra (2014) offer a more specific decomposition of a business model that applies to the manufacturing and trade industries. They argue that a business model can be segmented into how a "tangible product is marketed, delivered to customers, or enhanced with services" (p. 12). Like the Business Model Canvas, the decomposition of Netessine and Girotra (2014) contains an area that focuses on targeting and reaching customers (customer targeting) and another area that focuses on operations to get products to customers (product delivery). Their last component—how tangible products are "enhanced with services"

(support services)—incorporates the finance component within the Business Model Canvas. Whereas financing covers an important aspect of services that manufacturing and trade firms offer to customers—for example, subscriptions, leasing, or pay-per-use models (Guajardo 2018, Uppari et al. 2020, Ladas et al. 2021)—manufacturing firms may also offer other enhancing services, for example, to help monitor use or even predictive maintenance for those products (Chan et al. 2019). Using the decomposition by Netessine and Girotra (2014), a business method innovation in the manufacturing and trade industries can improve on an element of a business model in any of the three components: customer targeting, product delivery, and support services.

Examples of business method innovations in each of the three components can further illustrate the ways in which business method innovations can augment product sales. The first component, *customer targeting*, relates to innovations that improve how firms identify, target, or transact with their customers and develop new value propositions tailored to customers' specific needs. A patented business method innovation that falls into this category is Carrier Corporation's patent (US 7110966) for a business method designed to interactively elicit customers' needs and matching those needs to a suitable air conditioning product.

The second component, *product delivery*, relates to innovations in how products are delivered to customers. A business method innovation that falls in this category is Dell's patent (US 7756760) for a computer algorithm that helps determine inventory shortage costs across many buyers and supply lines in make-to-order systems. This business method innovation considers supply factors and customer demand factors in generating shortage costs and helps recommend parts to expedite.

Finally, the third component, *support services*, covers innovations in the "service bundle" designed to enhance the focal product's value. Such services can include financing arrangements (e.g., subscriptions, product warranties, or products sold "as a service") that help firms provide a continuous level of interaction and support to customers (Kim et al. 2007, Guajardo et al. 2012); services that help customers share their products to defray costs (Bellos et al. 2017, Abhishek et al. 2021); or services that help monitor/enhance the use of products. A business method innovation falling in this category is Xerox Corporation's patent (US 5745883) for a method that implements a way for electronic printing machines shared among users to be metered and billed based on job type and urgency considerations.

In sum, we present a typology of business method innovations in industries involved in physical product

manufacturing and delivery, based on which component of the business model (that is, customer targeting, product delivery, and support services) the business method aims to improve. In Section 4, we will examine how well business method innovations in these sectors fit within such a typology. In Section 6, we will compare the performative effects of innovating on multiple components as opposed to innovating on any single component of the business model (Rivette and Kline 2000, Osterwalder and Pigneur 2010, Girotra and Netessine 2013).

Valuing Business Method Innovation: A Literature Review

Is engaging in business method innovation activity likely to create value for firms in the manufacturing and trade sectors? The literature on strategic operations has discussed the importance of paying attention to the mechanisms of how products are marketed, delivered, and enhanced with services, both generally (Girotra and Netessine 2011, 2014; Ramdas et al. 2012; Kavadias et al. 2016; Plambeck and Ramdas 2020) and in specific settings (Visnjic and Van Looy 2013, Avci et al. 2015, Agrawal and Bellos 2017, Guajardo 2018, Marinesi et al. 2018, Adbi et al. 2020, Buell et al. 2021, Lee et al. 2020, Uppari et al. 2020, Abhishek et al. 2021). Our use of the term "business model," as concerned with the marketing, delivery, and services of tangible products, aligns with the definition of business model from this literature as applied to manufacturing and trade firms (for a conceptualization of the business model applied to service firms; Ramdas et al. 2012).

However, our work differs in that we focus on business method *innovation*—improving the components of the business model—rather than on how any specific business model can create value (Foss and Saebi 2017). Firms must push boundaries and develop a novel and nonobvious business method (i.e., business methods that are sufficiently different from existing innovations) as a condition of obtaining a patent. Because innovation is complex, uncertain, and disorderly, and because business method innovation is specifically about restructuring and reorganizing work within the organization, these activities are neither costless nor do they guarantee payoffs (Oliva and Kallenberg 2003, Sosna et al. 2010). Additionally, because developing new products and manufacturing processes are traditional sources of competitive advantage for manufacturers (Oliva and Kallenberg 2003, Suarez et al. 2013, Visnjic and Van Looy 2013), innovation activities relating to business methods can draw attention and resources away from a firm's core strength. Thus, business method innovation can be inherently risky (Massa and Tucci 2013), a sentiment

Johnson (2010) has echoed in contending that firms should focus on such innovations only after exhausting opportunities for innovations in products and manufacturing processes.

Therefore, although the empirical literature has found strong evidence of large and positive returns to product innovation (Sood and Tellis 2009), such conclusive evidence has not been established for business method innovation (Klingenberg et al. 2013). Related evidence, however, can be found in more general settings that study how management practices vary across firms (Bloom and Van Reenen 2007, Bloom et al. 2012, McKenzie and Woodruff 2017) and what impact good practices can have on firm value (Bloom et al. 2013, Anderson et al. 2018).

Two major differences emerge between the prior empirical work on management practices and the present study. First, although the stream of work by Bloom and colleagues (Bloom and Van Reenen 2007; Bloom et al. 2012, 2013) relates to firms' ability to deliver products (e.g., keeping inventory well organized), a much larger part of their study concerns the general management of the firm, such as setting targets and performance evaluations. Therefore, it is unclear which dimensions of management are leading to the significant gains seen, even in field experiment settings (Bloom et al. 2013). In comparison, the present study focuses solely on business methods aimed at supporting the sales of tangible products. Second, although this prior body of work focuses on whether firms are engaging in simple but widely adopted practices that are unlikely to incur significant costs (Bloom et al. 2013), the present work focuses on examining innovative methods that are not widely known—that is, those that are sufficiently novel and non-obvious to satisfy the requirements for obtaining a patent. To achieve such innovations, firms must incur the costs and risks associated with developing and launching new business methods. We therefore seek to understand if firms' engagement in business method innovation—creating novel improvements to aspects of their existing business model—generate value for the innovating firm.

The State Street Decision and Patenting Business Methods

Although some firms have always been innovative in their business practices, patenting business methods is a fairly recent phenomenon. The USPTO issues patents for any innovation that meets certain eligibility requirements. Such requirements include demonstrating that the invention is proper subject matter for a patent (e.g., a process, machine, manufacture, or composition of matter) and is novel and nonobvious to practitioners—that is, different enough from existing innovations to merit a patent (35 U.S.C. § 101-103).

Although business methods are technically “processes,” before 1998, business method innovations were

excluded from patentability on the grounds that they were not proper subject matter for patents (Merges 1999).² This changed in 1998 when the Court of Appeals for the Federal Circuit (CAFC) issued a decision in *State Street Bank and Trust Co. v. Signature Financial Group, Inc.* In upholding a business method patent for administering mutual funds, the CAFC held that methods of doing business were appropriate subject matter for patent protection if they produced a “useful, concrete and tangible result” (and provided they met other patenting requirements, e.g., being novel and nonobvious to practitioners). *State Street's* permissive stance toward business method patenting was unexpected (Jepsen 2006), and it led to a tidal wave of business method patent applications (Meurer 2002).

State Street's permissive stance toward business method patenting also triggered policy critiques (Burk and Lemley 2003, Bessen and Meurer 2006), which prompted courts and policymakers to introduce subsequent adjustments to the legal standards for obtaining and enforcing business method patents (Kasdan 2015). For example, a common critique is that business method patents can lack novelty and nonobviousness, despite being granted by the USPTO (Allison and Tiller 2003). Thus, soon after the *State Street* decision, Congress enacted the First Inventor Defense Act of 1999, which created a “prior user” defense specifically for business method patents. Alleged infringers were protected from liability provided they had used the business method commercially at least one year before the patent's filing date. Administratively, the USPTO has also singled out business method patents for additional quality control measures (Allison and Tiller 2003). Since the *State Street* decision was issued, subsequent court decisions have also adjusted the legal requirements for obtaining business method patents; for example, the Supreme Court's 2014 decision in *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014) raised the standard for demonstrating patentable subject matter.

Despite these later adjustments, *State Street's* recognition of business methods as patentable inventions created an unprecedented opportunity for firms to obtain such patents. A patent allows a firm exclusive use over the business method innovation it owns over the lifetime of the patent. This means that the firm can sue any entity that “makes,” “uses,” or “sells” the patented invention without a license from the patent holder (35 U.S.C. § 271). For example, Amazon sued Barnes and Noble for infringement of its “one-click” patent (which allows shoppers to store personal details on its website and make future purchases with a single click), while licensing the same patent to Apple for use in its online stores. In another example, Dell leveraged business method patents related to its build-to-order system in a \$16 billion cross-licensing deal

with IBM that yielded Dell lower-cost components (Rivette and Kline 2000). These examples suggest that business method innovations that are commercially useful—in that competitors want to imitate and deploy those approaches—have their value proportionately reflected in the ownership (Moore 2005, Kogan et al. 2017).

Importantly, *State Street* incentivized business method innovators to disclose such innovations as opposed to protecting them as trade secrets (Baird 2001). In so doing, *State Street* allowed the financial market to observe, analyze, and evaluate business method innovations in US firms. Such detailed insight was previously unavailable to the market. The market may assess firms with such innovations positively or negatively, depending on whether those innovations are perceived to help create or destroy firm value. The *State Street* event thus helped the market separate business method innovators from nonbusiness method innovators and allowed researchers to exploit the event to examine how the market's valuation of these two groups changed differentially over time.

Interestingly, almost all prior empirical studies of business method patents have focused on business methods generated by firms in the information and communications technology (ICT) industries (Allison and Tiller 2003, Hall and MacGarvie 2010) or by firms in the finance industry (Lerner 2002). In these industries, business method innovations are more akin to standalone software or finance product innovations—rather than innovations in the underlying business operations supporting the sales of tangible products. This is perhaps why business methods in the software and finance sectors are more likely to be trademarked—to add intellectual property protection and recognition to the brand or label associated with the method itself (Graham and Somaya 2004, Desyllas and Sako 2013). These studies, however, have overlooked an important subset of business method innovations—that of firms dealing with tangible goods. Notably, business method patents granted to firms in the manufacturing and trade sectors account for approximately 30% of all business method patents the USPTO has issued. Unlike ICT firms, business method innovations in the manufacturing and trade sectors primarily concern the business operations supporting the sales of physical products. Thus, they support and enhance the firm's business model. The present paper thus aims to identify whether engaging in such business method innovation activity—as measured through patents—creates value for firms in these sectors.

Methodologically, our work leans on approaches that have exploited legal changes (specifically, relating to intellectual property law) to deliver identification (Branstetter et al. 2006, Lerner 2009, Hall and

MacGarvie 2010, Bhaskarabhatla and Hegde 2014, Png 2017, Hegde and Luo 2018). Our work bears methodological similarity to Bhaskarabhatla and Hegde (2014), in using patents as a visible and measurable innovation output and in assessing the effect of a change in intellectual property law on firms' market value as measured by the Tobin's q ratio.

Business Method Patents in the United States: Data and Trends

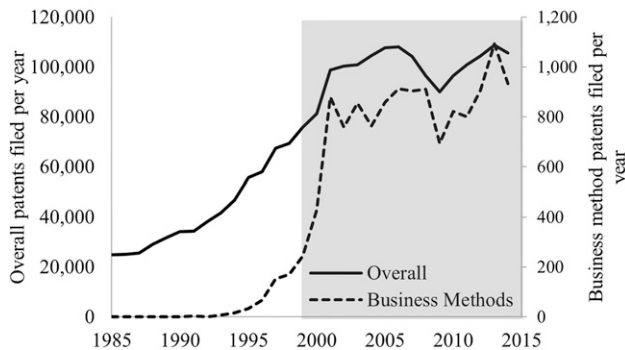
The USPTO categorizes business method innovations into "Class 705." The class definition describes "methods for performing data processing operations ... or calculation operations [that are] uniquely designed for or utilized in the practice, administration, or management of an enterprise, or in the processing of financial data, or determination of the charge for goods and services" (USPTO 2017). Innovations classified into 705 focus solely on innovating or improving business methods (Hall 2003). Although these innovations may rely on existing technologies in implementation (e.g., computers, the web, moving equipment), they do not innovate on those technologies.

To study business method innovations in the U.S. manufacturing and trade sectors, we first gathered all patent data during the period 1985–2014. This span avoided major legal changes in patent protection—specifically, the founding of the Court of Appeals for the Federal Circuit in 1982 (Bhaskarabhatla and Hegde 2014), and the U.S. Supreme Court's *Alice* decision in 2014, which placed additional restrictions on business method patenting.

We then gathered additional accounting and stock price data (at the firm-year level) from the database maintained by the Center for Research in Securities Prices (CRSP) and Compustat. We identified each firm's patenting activity by merging this data set with patent grant data that was publicly available from the USPTO. This procedure relied on the extensively curated firm identifiers from the NBER Patent Data Project (Hall et al. 2005), which covers patents granted from 1976 to 2006. For patent grants after 2007, we relied on the name-matching algorithm of Bessen (2008) to match firms between databases.

The merging of databases allows us to identify the patenting activities of firms in our sectors of interest: namely, public U.S. firms in the manufacturing (North American Industry Classification System NAICS 31–33) and trade—wholesale (NAICS 42), retail (NAICS 44–45), and transportation and warehousing (NAICS 48–49)—sectors. Figure 1 shows the growth in business method patent applications in these sectors during this period relative to overall patenting. Observe in Figure 1 that there were almost no business method patent applications before 1996, the year when the *State Street* case started moving

Figure 1. Annual Number of All Patents Filed (Solid Line) and Business Method Patents Filed (Dashed Line) in the Manufacturing and Trade Sectors Before and After *State Street*



Note. The shaded area shows the rapid acceleration in business method filings after *State Street* is decided in 1998.

through the court system. In the immediate aftermath of *State Street*, however, we observe a sharp jump in the number of business method patent applications, as firms in our sectors of study were quick to seize the unexpected opportunity to seek patents for their novel business methods (Meurer 2002).

However, the increase is attributable to a small subset of firms; in the 10 years after *State Street*, we see only 5% of U.S. firms in these sectors applied for and obtained business method patents. These observations suggest that the effect of *State Street* was mostly about incentivizing the subset of firms engaged in business method innovation pre *State Street* to now seek patents for their novel (or at least unique) business methods rather than encouraging firms not previously engaged in business method innovation to now do so. This is consistent with the empirical evidence in the innovation and patent literature showing that firms tend to seek patents where possible, especially firms in the sectors we studied (Moore 2005, Arora et al. 2016). Indeed, firms will not make changes to their innovation direction (which is costly) just to take advantage of patenting opportunities (Lerner 2009).

In sum, *State Street* allowed firms that were already innovating in business method innovation to begin disclosing and patenting those innovations. In doing so, these firms exposed the presence and content of their innovations, revealing information that affects how the financial market values them (Kogan et al. 2017). To better exploit the sharp nature of this unexpected event, we avoid using the entire 30 years of data we have, but instead focused on the 10-year periods before and after *State Street*. Thus, we identified 55,515 firm-year observations of publicly listed US manufacturers and trade firms for the years 1986–1995 and 1999–2008 (excluding 1996–1998, the years in which the *State Street* case was moving through the court system); we will describe this firm-year panel in

detail in Section 5, where we use it to establish the value of business method innovations. Over the period of 1986–2008 (inclusive of the transition years), we also identified 1,068 business method patents granted to firms in these sectors. We turn to examining the content of these business method innovations in the next section.

Analysis: Categorizing Business Method Innovations

Our first research goal is to understand the kinds of business method innovations that firms in the manufacturing and trade sectors pursue. Given that the term business model has been used to mean different things (Foss and Saebi 2017), we use categorization (Bowker and Star 1999) to identify the conceptual classes to which these patents belong. In doing so, we empirically examine how well business method innovations in these sectors fit with the conceptual categories of customer targeting, product delivery, and support services (Girotra and Netessine 2014).

Based on the function or industrial setting of the business method, the USPTO organizes different kinds of business methods into a set of hierarchically arranged subclasses—more than 40 main subclasses and about 250 fine subclasses. Following Hall et al. (2001), in Table 1, we first attempt to group the main subclasses into the three conceptual categories based on our own extensive reading of subclass descriptions and patent examples (see Online Appendix A for a sample of patents in each area).³

The first set of innovations in our sample (415 patents) are those involved with customer targeting. These subclasses encompass everything from electronic shopping/discounting/advertising to managing or securing customer transactions at the point-of-sale. We also include in this set subclasses related to innovations on how products are appraised, rated, or certified. These latter subclasses, together with others in Table 1 in italics, have small representations in our timeline of study (typically empty and less than four patents in each subclass).

To better understand the kinds of business method innovations in each category, we also parsed the patent documents to obtain the most important words that appear in the abstracts of the patents in the category. For example, the top keywords in the abstracts of patents in *customer targeting* put specific emphasis on words such as “digital,” “user,” “content,” “device,” “product,” “key,” “media,” “customer,” “transaction,” “consumer,” and “image” (as measured by tf-idf, “term-frequency inverse-document-frequency;” Jones 1972).⁴ Broadly, patents in this list cover innovations that involve customer targeting functions, typically before

Table 1. Categories of Business Method Patents (1,068 Patents)

Category	Main subclass descriptions	Subclass
Customer targeting (415 patents)	Electronic shopping; advertising; point of sale transactions; securing transactions; discount or incentives (coupon, rebate, offer, up sale); e-negotiation; <i>product appraisal; business/product ratings; product certification</i>	26.1; 14.1; 14.4; 16; 50; 80; 306; 317; 347
Product delivery (188 patents)	Operations research and analysis; inventory management; shipping; <i>reservation management; transportation facility access</i>	7.11; 28; 330; 5; 13
Support services (457 patents)	Costing/pricing; record and billing management; accounting; finance; insurance; customer communications; <i>warranty; product recall; after sales service; repair and maintenance admin; recycling or disposal; rental and leasing</i>	400; 2; 30; 35; 4; 346; 302; 303; 304; 305; 307; 308

Note. Main subclasses in italics have small representation (less than four patents) in our data.

sales are completed—for example, identifying relevant customers, gathering customer information, conveying product attributes, and closing and securing transactions.

A second set of innovations in our sample (188 patents) are those involved with *product delivery*. These innovations focus on how a firm organizes its operational activities and resources to facilitate product delivery—for example, how a firm coordinates equipment, shipments, schedules, and how it manages its finished and unfinished inventory. Here, we group patents in the subclasses of operations research and analysis, inventory management, shipping, reservation management, and transport facility access. The keyword list in the abstracts of this set of patents highlights words such as “inventory,” “data,” “resource,” “vehicle,” “customer,” “delivery,” “time,” “package,” “schedule,” “shipping,” and so on—items one would associate with activities facilitating physical product delivery.

The final category includes innovations in *support services*, covering 457 patents. These innovations typically focus on support activities after the product is delivered. In this category, we include subclasses that are involved in the costing and pricing of goods, finance, accounting, records and billing management, as well as managing after-sales services such as insurance, warranty, product recall, repair and maintenance, recycling, rental and leasing, and managing customer communications. Based on our reading of the patent documents, the innovations here focus on either how such support services should be metered and billed (supported by the presence of keywords related to “meter,” “automated,” “transaction,” “service,” “billing,” “payment,” “financial,” etc.) or how such services can help manage the product sold (e.g., tracking and administration of medication for patients—both “medication” and “patient” appear).

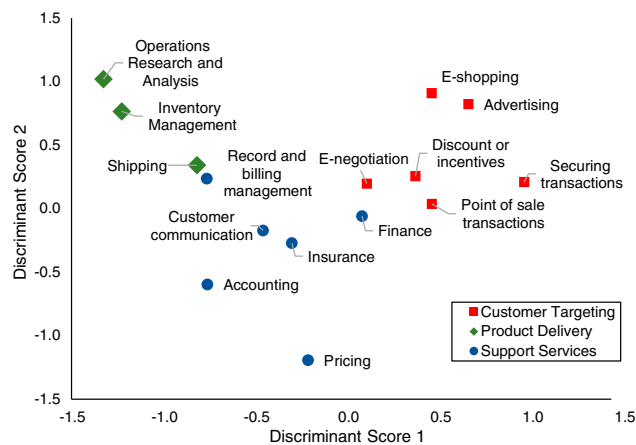
Validating the Categorization

To further validate the categorization described in the previous section, we report two analyses showing that (1) innovations within the same area of the business model are closer in meaning (relative to innovations that are in different areas) and (2) online observers, gathered from Amazon’s Mechanical Turk (MTurk), would categorize the subclasses in a similar way.

Our first validation asks the question: are innovations classified within the same area of the business model closer in meaning? One way to perform such an analysis is to compare patent abstracts to establish measures of similarity. Advances in machine learning of texts now enable such comparisons to be done at scale. Particularly, *Doc2Vec*, a state-of-the-art approach developed by Google (Le and Mikolov 2014), offers two advantages in such tasks: (1) analytically, it takes into account similarity in word meanings (e.g., “consumer” and “customer” are similar in meaning) when comparing documents, and (2) from an implementation standpoint, its popularity has encouraged many studies to use the approach, with guidance on the choice of parameters when running the model (Lau and Baldwin 2016).

First, we fed our data set of 1,060 patent abstracts into Gensim’s popular “Distributed-Bag-of-Words” implementation of *Doc2Vec* (Rehurek and Sojka 2010).⁵ The process transformed these 1,060 patents into 1,060 points in an $\mathbb{R}^{50}$ vector-space, whereby points with similar meanings are placed closer to each other.⁶ Next, to help us understand the distribution of patents across this space, we applied linear discriminant analysis to reduce $\mathbb{R}^{50}$ to $\mathbb{R}^2$. Intuitively, linear discriminant analysis finds two vectors in $\mathbb{R}^{50}$ vector-space, along which directions one can use to “best separate” the three areas of the business model indicated

Figure 2. (Color online) Average Location of Patents in Each Subclass, in the Directions of the Two Discriminant Vectors



Note. The two axes are not easily interpretable; however, higher distances between subclasses indicates greater dissimilarity between those subclasses.

in the previous section.⁷ Importantly, dimension reduction using linear discriminant analysis allows us to visualize the locations of the main subclasses in $\mathbb{R}^2$. We do so in Figure 2, where we plot the average location of patents in each of the 15 main subclasses with significant activity in our data (i.e., subclasses in Table 1 that are not italicized), along the directions of the two discriminant vectors.⁷ The plot indicates clear differences in the locations of subclasses (and therefore meanings) assigned to different components of the business model.

As a further analysis, we relied on MTurk to supply online observers that could help us categorize the same 15 main subclasses into the three areas of the business model (Kittur et al. 2008). In the survey, we first provide a definition of the three areas of the business model. We then present the title and description of a subclass, along with examples, asking how the respondent would classify the presented subclass into one of the three areas of the business model. Each respondent would iterate through and perform classification over all 15 subclasses, which are presented in random order. (Survey flow and results are in Online Appendix B.) We collected responses from 49 MTurk observers. Excluding five observers who failed verification checks, 44 observers furnished us with a total of 660 classification responses (44 observers $\times$ 15 each). We note that observers' responses agreed with our classification 65.5% of the time. This is far higher (at $p < 0.001$) than a null whereby observers randomly classify subclasses (where we would expect an agreement rate of 33.3%). Importantly, our classification *always* coincides with the modal, or most frequent, classification done by the MTurk observers.

Remarks on Classification

Overall, our content analysis reveals that most patented business method innovations in the manufacturing and trade sectors are concentrated in a few areas focusing on supporting the sales of physical products. They can be distinguished into three areas covering activities before and after sales: customer targeting, product delivery, and support services. Also informative are the kinds of business methods for which firms in the included industries *do not* typically innovate. A wide variety of main subclasses exist within class 705 that do not fit easily into the three categories (see Endnote 3). These include innovations in other business functions such as tax planning, intellectual property management, and legal services. Only eight patents from our set of business method innovations fall into these other subclasses.

Finally, although business method innovations are acknowledged as distinct from software innovations, the two are sometimes discussed together, because business method innovations are embedded predominantly in business software (Hall and MacGarvie 2010). Although most business method innovations lean on software, an important distinction exists between business method innovations and software innovations. A software product innovation culminates in a new computer program or software (Bessen and Hunt 2007). The novelty in business method innovations, however, is in the business method; it may rely on software but does not necessarily create a novel version of it. We assess the prevalence of software-based business methods following the approach of Arora et al. (2013) and Branstetter et al. (2019)⁸ and identify 186 innovations (of a total of 1,068 business method innovations) being labeled as software patents. Therefore, in the manufacturing and trade sectors, most business method innovations do not necessarily include novel software.

Analysis: The Value of Business Method Innovation

In this section, we use the firm-year panel data mentioned in Section 3, consisting of public U.S. firms in the manufacturing (NAICS 31–33) and trade—wholesale (NAICS 42), retail (NAICS 44–45) and transportation and warehousing (NAICS 48–49) sectors—in the 10-year period before and after *State Street*, to study if engagement in business method innovation creates value for firms in these sectors.

Variables

We constructed the following variables for our empirical analysis.

Market Value. Our dependent variable is the (log of) Tobin's q , or the ratio of the market value of a firm's assets to its book value. Tobin's q is a standard measure for assessing the over- or undervaluation of a company, and it has been used widely as a performance metric, especially on capturing the value of intangible investments, such as innovation (Hall and MacGarvie 2010). Such a measure is also forward-looking, enhancing our ability to capture the value implications of *State Street* in a timely manner using difference-in-differences (Bhaskarabhatla and Hegde 2014).

We use the approach of Chung and Pruitt (1994) to calculate Tobin's q . This approach, although approximate, only requires financial and accounting information available from the Compustat database and yet can explain at least 96.6% of the variability in Tobin's q obtained via the more theoretically correct model of Lindenberg and Ross (1981) (Bharadwaj et al. 1999). We take the log of Tobin's q as our dependent variable. The log form is common in innovation studies (Hall and MacGarvie 2010, Bhaskarabhatla and Hegde 2014, Branstetter et al. 2019); Hirsch and Seaks (1993) compare the log form of the q ratio against the original measure, finding the former "strongly preferred" (p. 381).

Business Method Innovation. We use this indicator variable to identify firms that show evidence of business method (BM) innovation activity. Therefore, for a firm that filed and was granted at least one business method patent in the study period, we set *HasBM* equal to one (and to zero otherwise). This indicator identifies the treatment (business method innovators) versus the control (nonbusiness method innovators) groups and is both fixed at the firm level and invariant over time.

Firm Characteristics. Our regressions also control for key firm characteristics. We use three sets of variables. The first reflects a firm's time-varying innovation capabilities; the second reflects the firm's strategic positioning; and the last captures the firm's size and returns to existing assets.

We model a firm's time-varying innovation capabilities with two variables—its overall knowledge stock and its research and development (R&D) investments, both of which affect the firm's ability to create value through innovation. We follow the approach of Hall et al. (2005), Hall and MacGarvie (2010), and Branstetter et al. (2019) to measure knowledge stock using a firm's past patenting activity. We also follow their assumption of a 15% depreciation rate for older patents over time. Then, we sum these patent amounts to yield a (properly depreciated) measure of knowledge stock (denoted by *LogPatentStk*).⁹ Importantly, the ability to leverage on knowledge is contingent on

R&D spending (Pennetier et al. 2019). We thus also measure a firm's time-varying R&D investments using its R&D intensity—the ratio of R&D expenditures to firm sales (*LogR&DIntensity*).

A second set of variables controls for variations in firms' strategic positioning in the market. Our set of variables here closely follows the work of Finkelstein and Hambrick (1990). Specifically, we have (1) plant, property, and equipment newness (net PP&E/gross PP&E; denoted *PP&ENewness*); (2) nonproduction overhead (SGA expenses/sales; denoted *LogOverhead*); (3) inventory turnover (inventories/sales; denoted *LogInventory*); and (4) financial leverage (debt/equity; denoted *LogFinLev*).¹⁰ These variables reflect the firm's key management levers and vary with the firm's product-service market choices.

Finally, we include the following characteristics to ensure that value is not confounded with effects because of firm size—the amount of physical assets (denoted *LogAssets*) and human assets (the number of employees; denoted *LogEmployees*) or from returns to existing assets—return to assets (net income/assets; ROA). In Online Appendix C, we provide a description of these variables (and, where applicable, a formula linking to Compustat items), along with summary statistics and raw correlations.

Empirical Model

Our empirical approach exploits the *State Street* decision, which, by allowing business method patenting, helped expose the presence and content of business method innovations in firms. As mentioned, the *State Street* decision was unanticipated (Jepsen 2006). To identify if BM innovators benefited more than non-BM innovators, we set up the following difference-in-differences (DID) analysis:

$$\log Q_{ij} = \beta_1 \text{HasBM}_i \times \text{Post}_j + \beta X + f_i + y_j + \varepsilon_{ij}. \quad (1)$$

In this analysis, $\log Q_{ij}$ is the log of the Tobin's q ratio of firm i in year j . HasBM_i is an indicator variable that takes the value of one if firm i is a BM innovator (and zero otherwise). Post_j is an indicator variable that takes the value of one if year j is after the *State Street* event (and zero otherwise). X is the vector of control variables described in the previous section, and β the vector of associated coefficients. f_i represents firm-level fixed effects, whereas y_j represents year-level fixed effects. Firm fixed effects absorb both cross-firm differences in valuation, as well as the noninteracted term HasBM_i that one typically sees in a DID setup. Similarly, year fixed effects absorb and model more finely the noninteracted dummy term Post_j .¹¹ Finally, we allow errors ε_{ij} to be correlated (i.e., clustered) across observations of the same firm.¹²

We are interested in the sign and the size of β_1 , which is the coefficient of the interacted term $HasBM_i \times Post_j$; β_1 tells us the direction and size of the bump in market value to BM innovators (relative to non-BM innovators) in the years after the *State Street* event, accounting for any existing differences between firms before the *State Street* event.

This setup exploits *State Street* as a shock that changed firms' incentives from keeping their business methods as closely guarded secrets to disclosing them. That is, our empirical design has the variable $HasBM_i \times Post_j$ that switches from zero to one for BM innovators immediately after *State Street*. A plausible alternative would be to have this variable switch from zero to one for BM innovators only when we also observe that the firm first filed for BM patents. Although we show in Online Appendix D that our results are robust to this alternative setup, we note that patent filing dates is a late proxy of *when* innovations appear in the public domain (and thus having an influence on firm value). Anecdotal and empirical evidence has shown that firms often publicize inventions before patents are filed (Kogan et al. 2017). Indeed, firms have the incentive to disclose patentable inventions as soon as they can, as such disclosures add to the body of evidence that a firm had been a "prior user" or "first inventor" of the invention (General Patent Corporation 2019). By leaning on *State Street* as an incentive shock toward disclosure, we provide a suitable alternative to the known problem of using patent filing dates to proxy the timing of disclosures of patentable inventions.

As a final note, identification of β_1 clearly comes from comparisons of firm values both before and after *State Street*. Companies can engage, however, in corporate actions (e.g., listings, mergers and acquisitions, spinoffs, and de-listings) that result in them entering or leaving the firm-year panel described in Section 3. Although unbalanced firm-year panels are common and do not threaten valuation analyses (Hendricks and Singhal 2001), firms that do not exist in either the pre- or post-*State Street* period do not directly contribute to the identification of β_1 . In the matching method described next, we will thus filter out firms that are only present in either the pre- or post-*State Street* period.

Matching

A crucial assumption of the DID analysis is comparability between the treatment group and the baseline group. Most importantly, the two sets of firms identified (BM innovators vs. non-BM innovators) should be comparable in that they exhibit parallel trends in the growth or decline of market value in the period before *State Street* (Angrist and Pischke 2009).

Here, we leverage a matching approach, specifically the "entropy balancing" matching method (Hainmueller 2012), to improve comparability of the sample in the pre-*State Street* period. To do so, we assign weights to firms in the control group (i.e., non-BM innovators) so that, after this assignment, the population of firms in the control group are *distributionally* similar (they have similar population means) to firms in the treatment group (i.e., BM innovators) regarding innovation and resource characteristics. In this sense, our approach is conceptually similar to other distribution matching approaches (e.g., "fine balance" matching as in Chan et al. 2019). It achieves covariate balance (similarity in the matched variables such as the firm operating and innovation variables). At the same time, it does so without omitting numerous observations, because the omission of large samples can potentially generate a larger estimation bias (Stuart 2010).

From an operational standpoint, we first aggregate all control variables (measured at the firm-year level) in the 1986–1995 pre-*State Street* period to create aggregate firm level measures. Then, we use the entropy balancing approach to weight firms in the baseline group (firms without evidence of BM innovations). By the end of the process, all firms in the treatment group (131 firms with BM innovation) stay in the sample and have a weight of one each. By contrast, firms in the baseline group are assigned weights equal to the total number of firms in the treatment group, so that we have 131 baseline firms that have means (across all control measures) that are similar to those of the treatment group.¹³

Table 2 shows the effectiveness of the matching process. Specifically, the first two columns compare the treatment and baseline samples *before* matching. We see large systematic differences—for example, firms in the baseline tend to be much smaller (in both assets and number of employees); have a much smaller patent stock but have higher R&D intensity and financial leverage (relative to size); and have a lower return on assets. The next two columns compare the treatment and baseline samples *after* matching. After matching, the population averages of the two sets of firms are now virtually the same.

Beyond similarity of firm characteristics across the groups, Table 3 reports tests of the parallel trends' assumption in the original and the matched samples. Here, using only pre-*State Street* data (1986–1995), the regression includes the interaction term $HasBM \times y_j$ (where y_j is the year-level fixed effects). In Model 1 (using the original sample), we see significant time-varying differences between the market value as measured by the log of Tobin's q (in this model, 1986—the first year of our sample—is set to be the baseline, where the aggregate difference in valuations between BM and non-BM innovators are set to zero).

Table 2. Comparison Between Treatment and Baseline Groups in the Original vs. Matched Samples

	Original sample		Matched sample	
	Treatment (<i>HasBM</i> = 1)	Baseline (<i>HasBM</i> = 0)	Treatment (<i>HasBM</i> = 1)	Baseline (<i>HasBM</i> = 0)
<i>LogPatentStk</i>	3.86 (2.37)	0.83 (1.30)	3.86 (2.37)	3.85 (2.33)
<i>LogRandDIntensity</i>	0.06 (0.07)	0.13 (0.42)	0.06 (0.07)	0.06 (0.22)
<i>PPandENewness</i>	0.57 (0.11)	0.59 (0.16)	0.57 (0.11)	0.57 (0.12)
<i>LogOverhead</i>	0.23 (0.15)	0.26 (0.31)	0.23 (0.15)	0.23 (0.31)
<i>LogInventory</i>	0.13 (0.07)	0.16 (0.15)	0.13 (0.07)	0.13 (0.07)
<i>LogFinLev</i>	0.39 (0.33)	0.83 (1.30)	0.39 (0.33)	0.39 (0.29)
<i>ROA</i>	0.06 (0.09)	−0.05 (0.34)	0.06 (0.09)	0.06 (0.08)
<i>LogAssets</i>	7.11 (2.14)	4.12 (1.84)	7.11 (2.14)	7.11 (2.04)
<i>LogEmployees</i>	2.50 (1.65)	0.78 (0.96)	2.50 (1.65)	2.49 (1.42)
<i>Industry</i>	—	—	Matched	Matched
<i>N</i> (firms)	131	2,419	131	131

Notes. The variables in this table represent firm-level averages for 1986–1995. The reported values are group means, with group standard deviations in parentheses. The table matches 131 treated firms to 131 (weighted) control firms from an original pool of 2,419 control firms.

Table 3. Analysis of the Parallel Trends’ Assumption in the Original vs. Matched Samples

	Model 1	Model 2
	Pre-State Street (original)	Pre-State Street (matched)
<i>HasBM</i> × 1987	0.09*** (0.02)	0.02 (0.03)
<i>HasBM</i> × 1988	0.07*** (0.02)	−0.01 (0.03)
<i>HasBM</i> × 1989	0.07*** (0.03)	−0.06 (0.04)
<i>HasBM</i> × 1990	0.14*** (0.03)	−0.05 (0.04)
<i>HasBM</i> × 1991	0.11*** (0.03)	−0.05 (0.04)
<i>HasBM</i> × 1992	0.12*** (0.03)	−0.04 (0.04)
<i>HasBM</i> × 1993	0.11*** (0.04)	−0.05 (0.04)
<i>HasBM</i> × 1994	0.22*** (0.04)	0.00 (0.05)
<i>HasBM</i> × 1995	0.24*** (0.04)	0.02 (0.05)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Within R^2	0.0019	0.0045
<i>F</i>	6.58***	1.32
<i>N</i> (firm-year)	28,219	2,254

Notes. Standard errors (in parentheses) are clustered by firm. *N* in the matched sample (Model 2) represents the sum of weights across all observations. Coefficients represent the difference between the market value of the treatment group (BM innovators) relative to the baseline group (non-BM innovators) at each year before *State Street*. Baseline year is set to 1986, where the difference between BM innovators and non-BM innovators is set to zero.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

However, when performing the same analysis on the matched sample (Model 2), *none* of the coefficients are now statistically significant. More importantly, the F statistic of this model, which tests the joint significance of all the coefficients presented in this model (excluding the firm and year fixed effects; see Endnote 11), is not significant ($F = 1.32, p = 0.22$). Thus, we can conclude that the matching process has successfully helped to construct comparable samples.

In sum, entropy matching helps us construct a weighted control set of firms that has covariates akin in distribution to the treatment firms (BM innovators), and more importantly, show parallel trends in the pre-*State Street* period. When these firm-level weights are applied to our firm-year panel data, we have a matched sample consisting of an unbalanced panel of 4,381 firm-year observations over 262 firms (131 BM innovators matched against 131 non-BM innovators). We present detailed distribution of the variables on this matched sample in Online Appendix C.

Value of Engaging in Business Method Innovation

We begin our analysis by examining the (differential) effects of the *State Street* event on BM innovators (versus non-BM innovators) using Equation (1) and the matched sample. The results are presented in Model 3 (Table 4).

The coefficient of $HasBM \times Post$ in Model 3 is $\beta_1 = 0.09$ ($p < 0.05$). Given the log scale of our dependent variable, the coefficient implies that BM innovators (firms with patented business methods) gained, on average, a 9% market value bump because of the *State Street* event, relative to non-BM innovators (firms without such patents). Thus, we find evidence that the market increased the valuation of BM innovators relative to non-BM innovators in the post-*State Street* period.

Our analysis, however, may be contaminated by temporal noise (e.g., firms changing strategic directions), that is, the observable part for which we account

Table 4. Effect of Business Method Innovations on Firms: GLS Model Predicting (log) Tobin's q

	Model 3	Model 4	Model 5
	Matched U.S. firms (1986–1995/1999–2008)	Matched U.S. firms (1991–1995/1999–2003)	U.S. BM firms matched to non-BM Europe firms
$HasBM \times Post$	0.09** (0.04)	0.08* (0.05)	0.14* (0.08)
Control variables			
$LogPatentStk$	−0.02 (0.02)	−0.01 (0.02)	−0.07** (0.03)
$LogRandDIntensity$	0.36*** (0.07)	0.41*** (0.14)	0.45*** (0.06)
$PPandENewness$	0.47*** (0.11)	0.37** (0.18)	0.68*** (0.17)
$LogOverhead$	−0.10 (0.16)	−0.08 (0.18)	0.43*** (0.16)
$LogInventory$	−0.58*** (0.20)	−0.14 (0.43)	−0.92** (0.23)
$LogFinLev$	−0.06*** (0.02)	−0.08** (0.04)	−0.08* (0.04)
ROA	0.78*** (0.14)	0.66*** (0.14)	0.08 (0.07)
$LogAssets$	−0.13*** (0.03)	−0.13*** (0.04)	−0.03 (0.04)
$LogEmployees$	0.05** (0.03)	0.08** (0.04)	0.03* (0.02)
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Within R^2	0.137	0.095	0.125
F	24.9***	11.8***	18.7***
N (firm-year)	4,381	1,936	3,245
Number of firms	262	216	202

Notes. Standard errors (in parentheses) are clustered by firm. N (firm-year) represents the sum of weights across all observations in the matched sample. “Number of firms” represents the number of firms that are matched successfully (i.e., the number of firms in the treatment sample plus the sum of firm weights in the control sample). Matching is done using all the control variables, and in the same way across all models.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

in the time-varying controls. We thus consider a model with shorter timeframes, where our results are less affected by such temporal changes. Thus, Model 4 focuses on just the five-year period before *State Street* went to court (1991–1995) and the five-year period after *State Street* was decided (1999–2003). Importantly, in doing so, we (1) reidentified firms with and without BM innovation based on the narrower window and (2) reran the matching process above (in the same way). Our results are consistent in this specification ($\beta_1 = 0.08, p < 0.10$).

Relatedly, we have argued that the baseline set of firms (non-BM innovators in the United States) are unaffected by the *State Street* event—or, they have no existing BM innovation capabilities to leverage the new patenting possibilities. We thus test whether our results are robust to an alternative baseline set of firms—European manufacturing and trade firms—who were unaffected by the event. The idea is that the *State Street* event was unique to the United States, and it marked a “clear divergence between the United States’ and European patent law” (Fink 2004, p. 299). Thus, European firms are valid as a control set because, regardless of whether they are innovators or not, European law has consistently prohibited the patenting of business methods (see the European Patent Convention of 1973, Article 52(2)(c)).

To maintain a clean comparison, we limited this control set to exclude European firms that either had a dual listing in any U.S. stock exchange or had any patents granted to them in class 705 (or the related but more restrictive International Patent Classification’s class G06Q used in Europe) in our period of study.¹⁴ We detail the construction of this set of European firms (which uses data from Compustat Global and the European Patent Office’s PATSTAT database) and sample characteristics in Online Appendix D. We test this in Model 5. To maintain comparability between the treatment set (United States-based BM innovators) and the control set (European firms), we estimated the model after conducting entropy matching (done in the same way as other models). Our results remain robust ($\beta_1 = 0.14, p < 0.10$).

Other threats to our model include the idea that the *State Street* event did not help BM innovators per se. Instead, what may have happened is that it benefitted manufacturers and distributors who have innovations in closely related areas, such as software or manufacturing processes. As discussed in Section 4.2, although business method innovations are acknowledged as distinct from software innovations, the two types are sometimes discussed as a pair, because business method innovations are typically embedded in business software (Hall and MacGarvie 2010). More generally, business methods also fall under the class of process patents, which includes manufacturing process innovations (35 U.S.C. 100(b)). To exclude these explanations, we ran

two placebo tests to show that the value effects of the *State Street* event were specific to BM innovators and did not create benefits for software and manufacturing process technology innovators (results are presented in Online Appendix D).

Finally, we also conducted the following robustness tests (presented in Online Appendix D): (1) a patent-level analysis where we show that business method patents are more likely to be renewed compared with other kinds of patents, which provides alternative evidence of the value of such innovations; (2) a Buy and Hold Returns analysis (Hendricks and Singhal 2001), in which we show that holding a portfolio of BM innovators through the post-*State Street* period would have produced higher returns than holding a portfolio of non-BM innovators with entropy matched characteristics; (3) an analysis based on alternative firm-patent matching data; and finally (4) using the filing date of the first BM patent of the firm as an indicator for the timing of a firm’s engagement with business method innovation. Our insights are robust to all these alternative approaches.

Additional Analysis: from Where Might the Value Emerge?

Thus far, we have presented evidence that *State Street* resulted in BM innovators in the manufacturing and trade sectors receiving a significant value bump relative to non-BM innovators. This additional value reflects the market’s evaluation of the firms’ business method innovations in the post-*State Street* era.

In this section, we dig deeper to discover which kind of BM innovators gained more from the ability to patent business methods. Our first analysis, presented in Table 5, splits our sample across manufacturers (Model 6) and other firms in the trade (wholesale/retail/transport and warehousing) sectors (Model 7), to examine the differential effects of the *State Street* shock. In Model 6, we see that BM innovators in the manufacturing sector gained about 7% in value after *State Street* (relative to non-BM innovators in the same sector; $\beta_1 = 0.07, p < 0.10$). In contrast, Model 7 illustrates that BM innovators in the other sectors gained a 25% in value after *State Street* (again, relative to non-BM innovators in these sectors; $\beta_1 = 0.25, p < 0.01$).

Because the core business of trade firms centers on customer targeting, product delivery, and providing support services, our result emphasizes the importance of innovating on elements of the core business. Importantly, the much smaller value bump we see for manufacturing firms is consistent with the idea that balancing product and business method innovation can be difficult. Nonetheless, our results suggest that even manufacturers should not turn a blind eye toward business method innovation.

Table 5. Heterogeneous Effects of Business Method Innovation: GLS Model Predicting (log) Tobin's q

	Model 6	Model 7	Model 8	Model 9
	Matched U.S. manufacturers (NAICS 31–33)	Matched U.S. firms in wholesale/retail/transport and warehousing (NAICS 42–49)	Effects of innovation scope (based on broad categories)	Effects of software-based BM innovations
<i>HasBM</i> × <i>Post</i>	0.07* (0.04)	0.25*** (0.06)		0.08** (0.04)
<i>HasCustomerTargeting</i> × <i>Post</i>			0.00 (0.06)	
<i>HasProductDelivery</i> × <i>Post</i>			0.02 (0.06)	
<i>HasSupportServices</i> × <i>Post</i>			0.04 (0.07)	
<i>Has2BMCategory</i> × <i>Post</i>			0.06 (0.06)	
<i>Has3BMCategory</i> × <i>Post</i>			0.18*** (0.06)	
<i>BMPatentCount</i> × <i>Post</i>			0.00 (0.00)	
<i>HasSoftwareBM</i> × <i>Post</i>				0.02 (0.05)
Control variables				
<i>LogPatentStk</i>	−0.02 (0.02)	−0.04 (0.04)	−0.02 (0.02)	−0.02 (0.02)
<i>LogRandDIntensity</i>	0.40*** (0.08)	0.86** (0.40)	0.36*** (0.07)	0.36*** (0.07)
<i>PPandENewness</i>	0.48*** (0.14)	0.55*** (0.15)	0.46*** (0.11)	0.47*** (0.11)
<i>LogOverhead</i>	0.04 (0.22)	−0.72** (0.30)	−0.10 (0.16)	−0.10 (0.17)
<i>LogInventory</i>	−0.66*** (0.16)	−0.21 (0.35)	−0.58*** (0.20)	−0.58*** (0.20)
<i>LogFinLev</i>	−0.06** (0.02)	−0.06* (0.03)	−0.06*** (0.02)	−0.06*** (0.02)
<i>ROA</i>	0.84*** (0.14)	0.50*** (0.13)	0.78*** (0.13)	0.78*** (0.14)
<i>LogAssets</i>	−0.13*** (0.03)	−0.10*** (0.03)	−0.13*** (0.03)	−0.13*** (0.03)
<i>LogEmployees</i>	0.05 (0.04)	0.10*** (0.03)	0.05** (0.03)	0.06* (0.03)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Within R^2	0.14	0.17	0.14	0.14
F	25.6***	7.2***	16.7***	22.9***
N (firm-year)	3,815	566	4,381	4,381
Number of firms	226	36	262	262

Notes. Standard errors (in parentheses) are clustered by firm. N in matched samples represents the sum of weights across all observations. The sample presented on this table are the same as the one in our main analysis (Model 3). In Models 6 and 7 we split the sample into manufacturers (Model 6) and others (Model 7).

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

In Model 8, we take a portfolio view and consider what kinds of business method innovations are worth pursuing. Both the general business model literature (Amit and Zott 2001, Massa and Tucci 2013) and the literature focusing on more specific definitions of “business model” involving the sales and delivery of tangible goods to which we adhere in this paper (Netessine and Girotra 2014) have proposed alternative ways of decomposing a business model into its

different components. However, all stress the importance of *alignment* between different components of the business model. The theoretical implication is that business method innovations in specific domains may not yield as large an advantage as innovations across multiple domains. Model 8 tests such predictions by examining the value implications of a BM innovator that is focused on only one component (i.e., innovations involved in only customer targeting, product

delivery, or support services), across two components, or across all three components of the business model. We do this by measuring BM innovation scope using a few indicator variables. Specifically, *HasCustomerTargeting*, *HasProductDelivery*, and *HasSupportServices* indicate that the firm has BM patents in only one identified component of the business model. Meanwhile, *Has2BMCategory* and *Has3BMCategory* indicate that the firm's BM patents cover two or three components. To ensure that the effects of economies of scope in which we are interested here are not confounded with the effects of economics of scale (i.e., increased valuation merely from having more BM innovations), we also include in this model *BMPatentCount* as a count of the number of BM patents granted to the firm over the timeline of our study.¹⁵

The results in Model 8 show that firms having only BM innovations in a single component do not see a value bump after *State Street* (the coefficients for *HasCustomerTargeting*, *HasProductDelivery*, and *HasSupportServices* are all not statistically significant). Firms having BM innovations covering two domains see a larger, but nonetheless not statistically significant, value effect of 0.06 ($p = 0.30$). Finally, we see that firms with BM innovations spanning all three components see a significant bump (effect size of 0.18; $p < 0.01$). Consistent with ideas from the business model literature, these results imply that innovation scope matters. That is, firms with BM innovations need to think about having business method innovations that span their entire business model.

On the other hand, we see little evidence of economies of scale. That is, the number of innovations seem to matter less than how extensively across the business model firms innovate (the coefficient of *BMPatentCount* $\times$ *Post* is not statistically significant; $p = 0.30$). Our results thus suggest that the marginal value of a patent is small, but the presence/absence of such innovations and where they are located has a much greater impact on value. This pattern of results is echoed in other contexts—for example, Hall and MacGarvie (2010) show that, although the marginal value of a software patent is insignificant to firms in the ICT sector, firms without any software patents nonetheless exhibit significantly lower market values.

Finally, in Model 9 we examine if BM innovations that are based on software—that is, the keywords “computer program” or “software” are sufficiently important to the innovation that it appears in the patent abstract (Bessen and Hunt 2007, Arora et al. 2013, Branstetter et al. 2019)—create more value than those that do not. We identify firms with such “software-based BM patents” with the indicator variable *HasSoftwareBM*. We do not find evidence that software-based business methods are more valuable ($p = 0.70$), while our results remain robust.

Discussion and Conclusion

Building directly on the perspective that business models reveal how an organization arranges its system of activities to create value (Zott and Amit 2010), the present study analyzes the value potential of innovations to the underlying business methods of established public firms. For firms in the manufacturing and trade sectors, these represent innovations in the activities, systems, and processes aimed at supporting the sales of tangible products (Netessine and Girotra 2014).

Scholars in operations management have offered conceptual arguments for why firms should be attentive to how their business models create and deliver value; they have also offered conceptual decompositions of the business model into areas where business method innovation can take place (Ramdas et al. 2012, Girotra and Netessine 2014, Plambeck and Ramdas 2020, Yakubovich and Wu 2022). Yet innovation in general is a costly endeavor, and business method innovation in established firms in particular can disrupt existing operations. Trying to deviate from established norms of doing business by no means guarantees success.

Leveraging on business method patents, which document firms' discrete attempts at changing the elements of their business model, we first performed a classification and textual analysis of the abstracts of the business method patents granted to companies in the manufacturing and trade sectors. Additionally, we validated our classification using both (1) a machine learning approach that maps patent abstracts to points in a “meaning space,” allowing us to show that our classification groups innovations closer in meaning to the same component of the business model; and (2) human online observers gathered from Amazon's Mechanical Turk, whose classifications were consistent with ours. Thus, our work empirically verifies that business method innovation in these sectors aligns with a three-part conceptual typology of the business model—how a firm targets customers, manages physical product delivery, and provide supports services (Girotra and Netessine 2014). In doing so, our work allows us to achieve conceptual clarity regarding what kinds of business method innovation occur in the manufacturing and trade context. Such conceptual clarity is useful for advancing research on the topic of business method innovation, where the term can mean different things in different settings (Lerner 2002, Bloom and Van Reenen 2007, Hall and MacGarvie 2010).

The present study also helps us understand how such innovations affect firm value. We show that firms that obtained business method patents gained a 9% bump in market value (as measured by Tobin's q) over comparable firms without any such patents in the post-*State Street* period. Because we compare business

method innovators and nonbusiness method-innovators, our results suggest that ignoring business method innovation entirely (and focusing only on other sorts of innovations) carry significant valuation penalties. In the decade following the *State Street* decision, which allowed the patenting of such innovations, only 5% of public firms in the manufacturing and trade sectors have applied for and received business method patents. This constitutes less than 1% of the patent portfolio of those firms. Thus, despite the growth in business method innovation in this sector, tremendous untapped opportunities remain.

We also provide more detailed insights into how business method innovations create value. First, we document that while manufacturers gain about a 7% bump in market value post-*State Street*, firms in the trade sectors (wholesale, retail, and transportation/warehousing) gain a 25% bump. The larger premium for firms in the trade sectors exemplifies an even stronger imperative to engage in business method innovation, as product or manufacturing process innovation opportunities do not typically compete for innovation resources within these companies. Meanwhile, the smaller bump for manufacturers suggests the difficulties of balancing different types of innovation activities (e.g., product innovation, manufacturing process innovation, and business method innovation). Nonetheless, our results suggest that manufacturers, too, should not ignore business method innovation.

Second, linking back to our classification work, we also show that firms focused on business method innovations that cover all three domains of the business model (customer targeting, product delivery, or support services) received a significant value bump relative to firms with business method innovations in only one or two domains. This finding is consistent with existing theories about business models (Chesbrough 2007, Netessine and Girotra 2014), which suggests that firms' ability to create and deliver value may depend not just on innovating specific and isolated business methods but also on creating alignment and cohesion among these innovations. While scholars have argued for the importance of alignment across different parts of the business model, we believe our work, by uniquely combining classification and valuation approaches, allows us to be the first to show this result empirically in a rigorous way. Our finding thus empirically validates the perspective of business models as a system of interdependent activities (Zott et al. 2011) and shows how improvements along the entire system of related activities (i.e., larger scope) provides greater value than innovations along any single dimension.

Limitations and Future Studies

We acknowledge the following limitations of the present research. First, our decomposition of the business

model into customer targeting, product delivery, and support services are specific to firms in the manufacturing and trade industries. Further conceptual and empirical work can provide a framework for such decompositions in the broader service industry (Ramdas et al. 2012). Our classification is also based mainly on the patent abstracts; going beyond the inventions' summary texts may reveal deeper insights on their contents. Nonetheless, we hope our framework—based on an initial classification validated further using a mix of machine-learning and human-assisted classification techniques—could serve as a guide toward future empirical efforts.

Indeed, we hope our classification approach can help policymakers make finer distinctions between different kinds of business method innovations, which may allow for more nuanced policymaking in the area of business method patents. Legal rulings that came after *State Street*, such as the U.S. Supreme Court's ruling in *Alice Corp. v. CLS Bank International* in 2014, have made it more difficult to patent business methods. Although many of these court decisions have involved business method patents on financial innovations, the resulting restrictions have broadly affected all kinds of business method innovations. Our work highlights the heterogeneity of business method innovations across different industry contexts and the benefits of developing a more finetuned understanding of different kinds of business method innovations.

Second, our identification is based on comparing firms with business method innovation matched against those without, but otherwise sharing similar operating/innovative characteristics. Firms in this set tend to be larger, more innovative, and produce a higher return on assets. We cannot generalize our insights to smaller, less innovative, or less profitable firms. Such firms face different sets of challenges (Massa and Tucci 2013). On the other hand, because they are less encumbered by existing business models, such firms may have opportunities to engage in business method innovation to generate phenomenal growth (Kavadias et al. 2016). Studying the conditions under which firms can leverage business method innovations to generate growth thus remains a promising avenue of future research (Massa and Tucci 2013).

Despite these limitations, our work can inform managerial decision-making in the manufacturing and trade sectors. Our work can also serve as a guide to other scholars on the usefulness of business method patents as an empirical basis for studying how firms innovate to improve their performance.

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Endnotes

¹ The term “business model” has been used in different contexts to mean different things (Foss and Saebi 2017). Our definition, focusing on how tangible products are marketed, delivered, or supported with services, is specific to firms in manufacturing and trade and aligns with how the term is used in the operations management literature.

² Despite this formal stance, the USPTO granted a small number of such patents prior to 1998 (USPTO 2017).

³ See <https://www.uspto.gov/web/patents/classification/uspc705/sched705.htm> (retrieved January 8, 2021) for subclass definitions. Many subclasses within 705 do not fit into the categories in Table 1. However, only eight patents in our sample fall into the “Other” category of subclasses: Voting or election arrangement; Restaurants; Tax strategies; Collaborative product development; Project management; Dispute resolution; IP management; Legal Services; Real estate; Social network; Human resources; Destination assistance; Personal security, identity or safety; Education services; Fundraising; Business/ product documentation; Employee communications; Simulation of business models; and Miscellaneous.

⁴ Intuitively, we create a measure for each word, which captures the propensity of it appearing in a document (tf), while discounting its propensity of appearing in general use (idf , e.g., the word “the” appears frequently but it is not informative). tf is operationalized by the number of occurrences of the focal word divided by the total length of the patent's title and abstract combined; idf is operationalized by the (inverse of) log of the number of documents containing the word. We present the terms with the highest tf - idf scores in each category of business method patents.

⁵ We exclude the eight patents in the “other” categories in this process. *Doc2Vec* only needs an identifier for the patent along with patent abstracts as inputs; we do not provide it with any other information.

⁶ The main decision points of using *Doc2Vec* are (1) choosing between “Distributed Memory” or “Distributed Bag of Words” implementation—there is evidence that the latter works better (Lau and Baldwin 2016); (2) the dimensions of the vector space—large datasets (like the entire collection of English Wikipedia or Google News articles) typically use models with 100–300 dimensions (Mikolov et al. 2013). Because our data set is much smaller, we follow the example of Rehurek (2021) of 50 dimensions for a corpus similar in size to ours; and (3) the size of the “context window”, or words around a focal word in a sentence that can be considered in the same context, here we use the default window of plus/minus five words in Gensim's *doc2vec* implementation (Le and Mikolov 2014).

⁷ Consider the problem of separating two sets of points X and Y ; the analysis then involves finding a vector v_1 and a constant c whereby the likelihood of having $v_1 \cdot x > c$ for points $x \in X$ and $v_1 \cdot y \leq c$ for points $y \in Y$ is maximized. For three sets of points (corresponding to the three areas of the business model), we would need two vectors v_1 and v_2 to be able to make such separations. Such a separation is not guaranteed. Indeed, both vectors were successful in, and are required to, discriminate between the three areas (Wilk's lambda = 0.67; $F(49, 1009) = 9.96, p < 0.001$).

⁸ In particular, they combine two techniques established in the literature: (1) adopting a keyword approach (e.g., “computer program” or “software”), while scanning the title and abstract of patents (Besen and Hunt 2007); and (2) labeling patents using “software” categories as defined by Graham and Mowery (2003) with the International Patent Classification system as a basis. We created an indicator variable *Software* that was set to one if a patent was declared a software patent under either technique (or set to zero otherwise).

⁹ We use a “log + 1” formulation for patent stock and R&D intensity to avoid taking the log of zero. All variables that are logged are prefixed by “Log.”

¹⁰ Finkelstein and Hambrick (1990) also include “R&D intensity” as a strategic lever, which we also do to model the firm's innovation capability. We exclude advertising spending, as it is part of the more widely reported “SGA expense” item.

¹¹ The fixed effects, f_i and y_j , do not need to be estimated if they are not of interest. The fixed effects estimator first transforms the data using a within-transformation to remove their effects, before estimating the remaining coefficients (Greene 2008, p. 192). We follow the within-transformation approach, both because we do not theorize their effects and to allow measures of “model-fit,” such as the within R^2 and the F statistic, to be predicated only on variables with which we are concerned (i.e., excludes f_i and y_j ; StataCorp LLC 2021).

¹² Clustered errors relax the OLS assumption that error terms are all independent, by allowing observations within the same firm panel to exhibit any form of correlation and is a standard relaxation of the “independent errors” assumption in OLS with fixed effects models (Cameron and Trivedi 2005). Relaxing the independent errors assumption also led to the model becoming “generalized” (thus GLS as opposed to OLS).

¹³ Weights are not limited to be integers, so a “baseline firm” can be composed of multiple firms with fractional weights. All subsequent regressions will account for observation weights in their estimation.

¹⁴ Europe has allowed limited patenting of business method innovations exhibiting technical character—that is, those containing novel hardware, software, or both and therefore more akin to technological inventions (Marsnik and Thomas 2011). These are typically housed in G06Q (Wagner 2008).

¹⁵ We leave out the eight patents in the “Others” category (see Endnote 3) for this analysis. Patents in subclass “record and billing management” are close in meaning to “shipping” but placed under support services (Figure 2). Our results are robust if we move them to product delivery. Consistent with our main approach of identifying BM innovators, these variables are all fixed at the firm level and time-invariant.

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